

Flink SQL reports

Seven reports that read two streams of isotope metadata — the **produce side** (the `x-isotope-*` headers stamped on every event topic record by `IsotopeProducerInterceptor`) and the **consume side** (the value-less marker records on `isotope_consume_edge_markers` emitted by `IsotopeContext.recordConsume`) — and surface what's flowing where, how fast, and how reliably. Each report runs as a long-lived `INSERT INTO <report>_1m SELECT ...` streaming job. The aggregation logic is identical across runtimes; only the source/sink DDL and function-registration glue differ.

The headline addition is the `bipartite_topology` report: it unions the produce and consume views to render the pipeline as a literal **bipartite graph** from graph theory — services in one vertex set, topics in the other, edges crossing between them in both directions. See [root README §1.0 "Background — why 'bipartite'?"](#) for the full motivation.

Runtime	Reports	Sink format	Where the SQL lives
Confluent Platform Flink (cp-flink 2.1.2 session cluster on Minikube)	7 (latency, topology, bipartite-topology, hop-distribution, coverage, stuck-trace, latency-percentiles)	<code>avro-confluent</code> (SR-framed Avro)	<code>scripts/flink/sql/cp/</code> — applied by <code>scripts/deploy-cp-flink-reports.sh</code>
Confluent Cloud for Apache Flink (CCAF)	7 (same set as CP)	<code>proto-registry</code> (SR-framed Protobuf)	Inlined as <code>confluent_flink_statement</code> resources in <code>terraform/setup-confluent-flink.tf</code> — applied by <code>scripts/deploy-cc-flink-reports.sh</code>

Control Center deserializes both sink formats natively.

What each report computes

Every report aggregates over a `TUMBLE(event_time, INTERVAL '1' MINUTE)` window. The "Source view" column names the typed view (or views) the report reads from; the views in turn filter the single `isotope_raw` source table by the presence/absence of the `x-isotope-consumer-service` header.

Every report also carries a `pipeline` column (decoded from the `x-isotope-pipeline` header) as a leading grouping dimension, so rows for an `orders` pipeline and a `location` pipeline never aggregate together. The "What it computes" column below lists the *additional* keys each report groups by, on top of `pipeline`.

Report	Source view	What it computes	Runtimes
latency	isotope	avg / min / max end-to-end latency by <code>origin_service</code> × <code>current_topic</code>	CP, CCAF
topology	isotope	produce-edge counts per (<code>producer_service</code> , <code>topic</code>) per minute	CP, CCAF
bipartite_topology	isotope and <code>consume_events</code>	full bipartite graph: produce edges plus consume edges per minute	CP, CCAF
hop_distribution	isotope	record counts bucketed by <code>hop_count</code> per topic per minute	CP, CCAF
coverage	isotope	distinct traces per topic per minute	CP, CCAF
stuck_trace	isotope	alerts (via <code>STUCK_TRACE_PTF</code>) for traces idle ≥60s of event time	CP, CCAF
latency_percentiles	isotope	p50 / p95 / p99 (via <code>LATENCY_PERCENTILES</code> PTF, T-Digest)	CP, CCAF

Format-by-runtime, not by domain

The sink **format** differs by runtime for one platform-level reason:

- **cp-flink ships SR-Avro but not SR-Protobuf.** Apache Flink open-source publishes `flink-sql-avro-confluent-registry`; there is no SR-integrated Protobuf equivalent. We could hand-write one (~150 lines wrapping `flink-protobuf` with magic-byte framing
 - SR client) but Avro already gives us Control Center decoding with zero custom code. CCAF has SR-Protobuf, so its sinks use it.

The report **set** is identical on both runtimes — but that took a deliberate choice for percentiles: **CCAF rejects all user-defined aggregate functions** with `aggregate functions are not supported`, regardless of accumulator shape. So `LATENCY_PERCENTILES` is implemented as a `ProcessTableFunction` (T-Digest sketch + per-window state and timers), not an aggregate function — a PTF registers and runs on both runtimes, same as `STUCK_TRACE_PTF`.

The demo *event* topics (`orders.placed`, `orders.enriched`, `orders.fulfilled`) ride Protobuf+SR via the Java app's `DemoEvent` schema on both runtimes — those are written by the Kafka producer client, not by Flink, so the SR-Protobuf gap doesn't apply.

Layout

```
scripts/flink/sql/cp/
  00_source_table.fql
= 'kafka') + isotope_raw UNION view
  01_register_functions.fql
```

CP Flink – session-cluster SQL
CREATE TABLE per topic ('connector'
CREATE FUNCTION ... USING JAR

```
'file:///opt/flink/lib/isotope-flink-udf.jar'
 05_isotope_view.fql          Typed view; decodes x-isotope-*
header scalars (produces only)
 06_consume_events_view.fql    Typed view of
isotope_consume_edge_markers markers (consume edges)
 05_report_sinks.fql          CREATE TABLE for each
isotope_report_*_1m Kafka sink (avro-confluent)
 10_latency_report.fql        INSERT INTO: avg/min/max latency by
origin x topic
 20_topology_report.fql        INSERT INTO: produce-edge counts
per minute
 25_bipartite_topology_report.fql INSERT INTO: produce plus consume
edges (bipartite graph) per minute
 30_hop_distribution.fql       INSERT INTO: hop-count buckets per
topic per minute
 40_coverage_report.fql        INSERT INTO: distinct traces per
topic per minute
 60_stuck_trace_report.fql     INSERT INTO: stuck-trace alerts via
STUCK_TRACE_PTF
 70_latency_percentiles_report.fql INSERT INTO: p50/p95/p99 via
LATENCY_PERCENTILES PTF (T-Digest)
 99_teardown.fql              DROP TABLE/VIEW/FUNCTION (companion
to flink-reports-down)
```

CCAF runs the same seven reports, but the SQL lives inline as `confluent_flink_statement` resources in [terraform/setup-confluent-flink.tf](#) — 23 statements: ALTER (×4) + raw view + typed produce view + typed consume view + 7 sinks + `STUCK_TRACE_PTF` and `LATENCY_PERCENTILES` registrations (×2) + 7 INSERTs. The JAR is uploaded as a `confluent_flink_artifact` and referenced via `USING JAR 'confluent-artifact://<id>'`.

Wire-format detail (CP only)

Window columns ride as `BIGINT` epoch millis on the wire, not `TIMESTAMP_LTZ`. Flink 2.1.2's `avro-confluent` schema-derivation path raises `UnsupportedOperationException: Unsupported to derive Schema for type: TIMESTAMP_LTZ(3)` for that type. We cast in the INSERT `(UNIX_TIMESTAMP(CAST(window_start AS STRING)) * 1000)` so the on-wire schema is plain Avro `long`. Consumers rehydrate via `TO_TIMESTAMP_LTZ(window_start, 3)`. The CCAF Protobuf sinks don't hit this — `proto-registry` handles `TIMESTAMP_LTZ` directly.

PTF JAR

The single shadow JAR `ptf/build/libs/isotope-flink-udf.jar` (produced by `./gradlew :ptf:shadowJar`) carries both JAR-backed functions — both `ProcessTableFunctions` — under the `ai.signalroom.kafka.isotope.flink` package:

- `LatencyPercentilesPTF` (registered as `LATENCY_PERCENTILES`) — T-Digest p50/p95/p99 via per-window state + event-time timers.
- `StuckTracePTF` — per-trace state + event-time timer that emits alerts for traces idle ≥60s.

Both register on both runtimes; only the `CREATE FUNCTION ... USING JAR ...` clause differs (`file://` path on CP, `confluent-artifact://` reference on CCAF).

Operations

CP Flink (Minikube):

```
make flink-up          # cert-manager → CFK Flink Operator → CMF → session
                        cluster
make flink-reports-up  # build PTF JAR, copy to JM, create sink topics,
                        submit 7 INSERT INTO jobs
make flink-sql         # interactive SQL Client (auto-loads sink DDL so
                        SELECT * works)
make flink-reports-down # cancel jobs, drop tables, delete sink topics + SR
                        subjects
make flink-down        # tear down cluster + operator + cert-manager
```

CCAF (Confluent Cloud, Terraform-driven):

```
make cc-flink-reports-up  CONFLUENT_API_KEY=... CONFLUENT_API_SECRET=...
                        # terraform apply: env + cluster + topics +
compute pool + artifact + 23 statements
                        # also regenerates terraform/terraform.png via
`terraform graph | dot`
source scripts/cc-cli-env.sh          # exports BOOTSTRAP / SR_URL /
KAFKA_KEY / KAFKA_SECRET / SR_KEY / SR_SECRET / JAAS
scripts/cc-app-run.sh send orders.placed order-intake-service 'hello' #
drives traffic with the SASL config
make cc-flink-reports-down CONFLUENT_API_KEY=... CONFLUENT_API_SECRET=...
                        # terraform destroy: deletes the environment and
                        everything in it
```

See the [root README §4.5](#) for the full CCAF walkthrough, including the multi-window sustained-traffic pattern required to see tumbling-window aggregates emit.